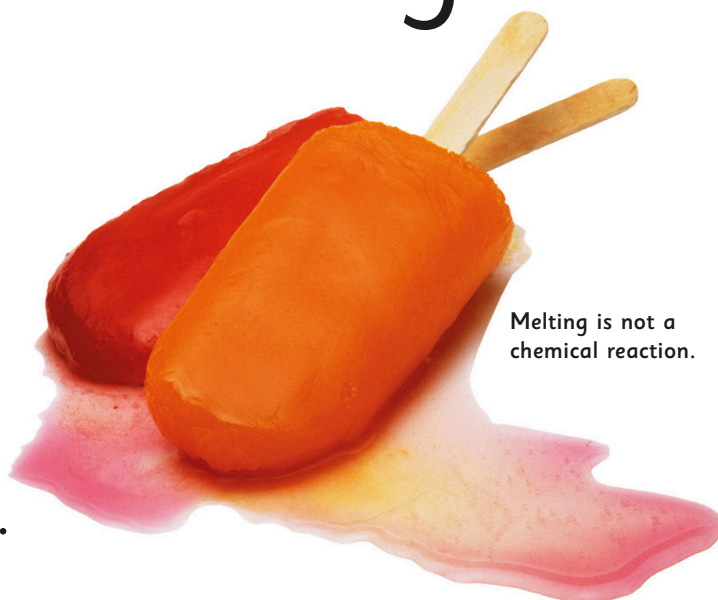


Reactions and changes

Materials change as a result of physical processes or chemical reactions. In a chemical reaction, atoms join with or break away from other atoms, forming different compounds. Chemical reactions often lead to a dramatic change.



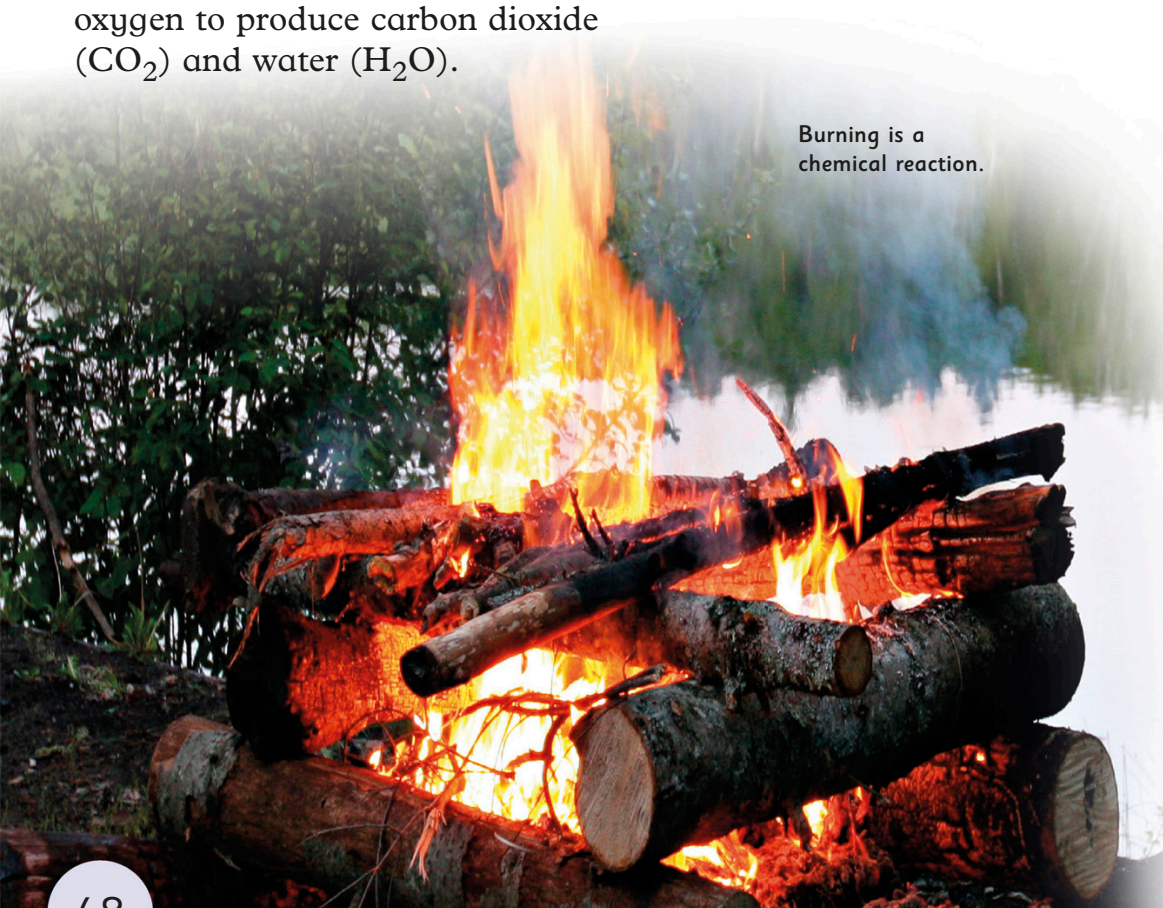
Melting is not a chemical reaction.

Chemical change

Burning is a chemical reaction involving oxygen (O). Wood is made of compounds containing carbon (C) and hydrogen (H). When it burns, the carbon and hydrogen react with oxygen to produce carbon dioxide (CO_2) and water (H_2O).

Physical change

Not all dramatic changes are caused by chemical reactions. When ice pops melt, the atoms in the water molecules do not get rearranged into new molecules—they remain water molecules. Melting is simply a physical change.



Burning is a chemical reaction.



Escaping energy

Chemical reactions can release energy as heat and light. A sparkler contains chemicals that release a lot of energy as light to create a dazzling shower of sparks.